DEVOPS FUNDAMENTALS PROJECT REPORT

Cloud-Based Linux Server Deployment with GitHub Version Control and Automation

Project area	Cloud infrastructure, Linux administration, web hosting, automation
Platform	Amazon Web Services (AWS) EC2 with Ubuntu Linux
Web server	Apache HTTP Server
Version control	Git and GitHub
Repository	github.com/vaishnavipable/DevOps-Fundamentals-Project

Prepared as a summary of the implementation and learning outcomes of the DevOps Fundamentals project.

1. Project Overview

This project demonstrates a practical workflow for deploying and managing a cloud-based Linux web server. It combines AWS EC2 infrastructure, Ubuntu administration, Apache website hosting, Git/GitHub version control, Bash automation, website backups, and Linux log analysis.

The goal is to practise core DevOps tasks: provision a server, publish a website, automate routine checks, keep project files under version control, inspect system logs, and document the results.

2. Objectives

- Provision and access an Ubuntu EC2 instance on AWS.
- Install and verify the Apache HTTP Server.
- Create and serve a custom website from the project's index.html file.
- Use Git and GitHub to version and publish project files.
- Develop Bash scripts for server health checks, Apache monitoring, and website backups.
- Review authentication and system logs and save findings in log-report.txt.
- Maintain project documentation and prepare a final report.

3. Architecture and Tools

The project uses an AWS EC2 instance running Ubuntu Linux. SSH provides administrative access to the instance. Apache serves the website, while Git tracks changes to the project repository and GitHub stores the remote copy.

Component	Role
AWS EC2	Hosts the Ubuntu server.
Ubuntu Linux	Operating system and command-line administration environment.
Security group	Controls network traffic according to the rules configured in AWS.
Apache HTTP Server	Serves the project website.
Git and GitHub	Track versions and store project files remotely.
Bash scripts	Automate server checks, Apache monitoring, and backups.

4. Implementation Summary

4.1 Server and website

An Ubuntu EC2 instance was configured for remote administration. Apache was installed and used to serve the custom website stored in index.html. Service status and local HTTP responses were checked during implementation.

4.2 Version control

The project is maintained in a Git repository and pushed to GitHub. Changes to scripts, backups, the log report, and README documentation were committed to the main branch.

Repository: https://github.com/vaishnavipable/DevOps-Fundamentals-Project

4.3 Automation scripts

Three executable Bash scripts are present in the scripts/ directory:

- **health-check.sh** — displays system information such as hostname, uptime, CPU, disk and memory usage, logged-in users, running processes, and Apache status.
- **apache-monitor.sh** — checks whether Apache is running and attempts to start it if it is stopped.
- **backup.sh** — creates a timestamped compressed archive of website files from /var/www/html and stores the archive in the Backup/ directory.

4.4 Website backup

The backup script was corrected so that it archives the website files in /var/www/html, rather than only index.html. Timestamped .tar.gz archives are stored in Backup/ to distinguish backup runs.

4.5 Linux log analysis

The log-report.txt file records the commands used and summarizes the selected entries from /var/log/auth.log and /var/log/syslog. The reviewed authentication entries included successful SSH public-key logins for the ubuntu account. No failed or invalid-user entries appeared in the displayed search results.

The reviewed system logs also contained recurring Amazon Systems Manager Agent messages about missing instance-role credentials and an AccessDeniedException indicating that the Systems Manager instance management role was not configured for the AWS account. A fwupd message also reported that /sys/power/mem_sleep was unavailable. These observations are limited to the selected log entries and do not, by themselves, establish an impact on the website.

5. Repository Structure

```
DevOps-Fundamentals-Project/
■■■ Backup/
■   ■■■ Timestamped website backup archives (.tar.gz)
■■■ docs/
■   ■■■ project-documentation.md
■■■ screenshots/
■   ■■■ health-check-output.png
■   ■■■ health-check-script.png
■   ■■■ project-files.png
■■■ scripts/
■   ■■■ apache-monitor.sh
■   ■■■ backup.sh
■   ■■■ health-check.sh
■■■ index.html
■■■ log-report.txt
■■■ README.md
```

The screenshots listed above are the files currently present in the repository's screenshots/ directory. Additional evidence may be kept separately on the project computer.

6. Commands Used

System administration and service checks

```
pwd
ls -la
sudo apt update
sudo systemctl status apache2
df -h
curl http://localhost
```

Git version control

```
git status
git add <file>
git commit -m "Describe the change"
git push origin main
```

Log review

```
sudo grep -i "failed" /var/log/auth.log | tail -20
sudo grep -iE "Accepted|Failed|Invalid user" /var/log/auth.log | tail -20
sudo grep -iE "error|critical|alert|emerg" /var/log/syslog | tail -20
sudo grep -i "warn" /var/log/syslog | tail -20
```

7. Challenges and Troubleshooting

- Establishing SSH access to the Ubuntu EC2 instance.
- Installing and verifying Apache and confirming that the website could be served.
- Configuring Git/GitHub version control and pushing project updates.
- Correcting the backup script to include all website files and use timestamped compressed archives.
- Interpreting system log messages, including AWS Systems Manager role and credential warnings.
- Keeping scripts, reports, backups, and documentation organized in the repository.

8. Learning Outcomes

- Basic cloud infrastructure and EC2 instance administration.
- Ubuntu Linux command-line operations and service management.
- Apache web server setup and website deployment.
- Git commits, repository status checks, and GitHub pushes.
- Bash scripting for repeatable monitoring and backup tasks.
- Compressed archive creation and timestamped backup management.
- Searching authentication and system logs and documenting observations.
- Technical writing and project documentation.

9. Deliverables and Verification

- GitHub repository with source files, scripts, and documentation.
- Three executable Bash scripts in scripts/.
- Timestamped website backup archive(s) in Backup/.
- Linux log analysis report: log-report.txt.
- README.md and docs/project-documentation.md.
- Evidence screenshots, including the files currently in screenshots/.
- This Project Report PDF.

Before final submission, verify that any additional screenshots required by the assignment—particularly AWS infrastructure and website evidence—are available and included if required by the instructor.

10. Conclusion

This project provided hands-on practice with a cloud-hosted Linux server and a basic DevOps workflow. The implementation brought together server setup, website hosting, version control, Bash automation, timestamped backups, and system log review. The repository and accompanying documentation provide a record of the work completed and the main lessons learned.